Applied Data Science Portfolio: Data Hamster

- La Monte Henry Piggy Yarroll
- Syracuse University IST782, Spring 2026

Peter Maas, CC BY-SA 2.5 <https://creativecommons.org/licenses/by-sa/2.5>, via Wikimedia Commons

- Name Navigator
- Sugar Bowl (Glycemic Increment)
- Synoptic Key of Life (SKOL)

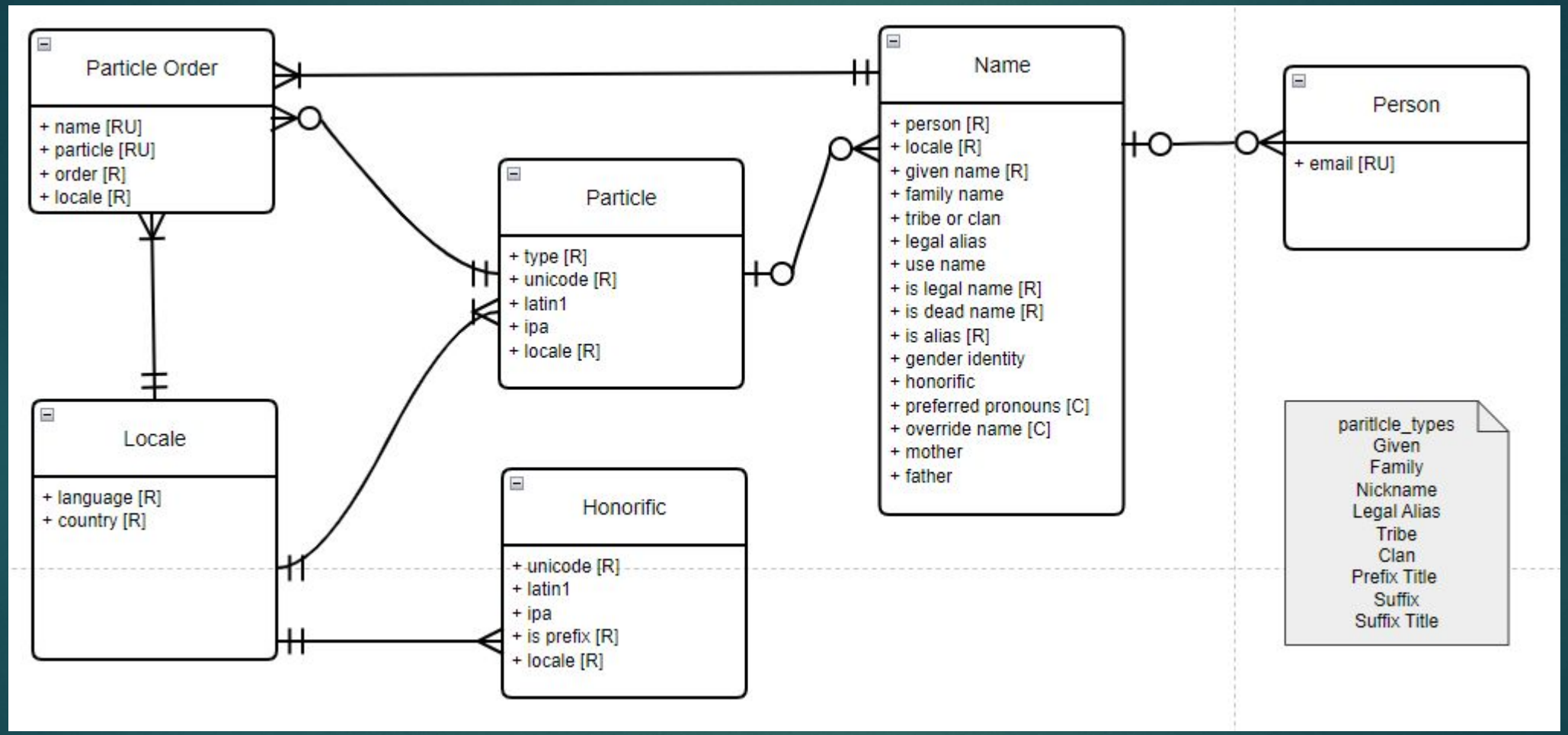
Name Navigator

Our goal is to achieve cultural awareness by accurately capturing and respecting the nuances of sensitive names.

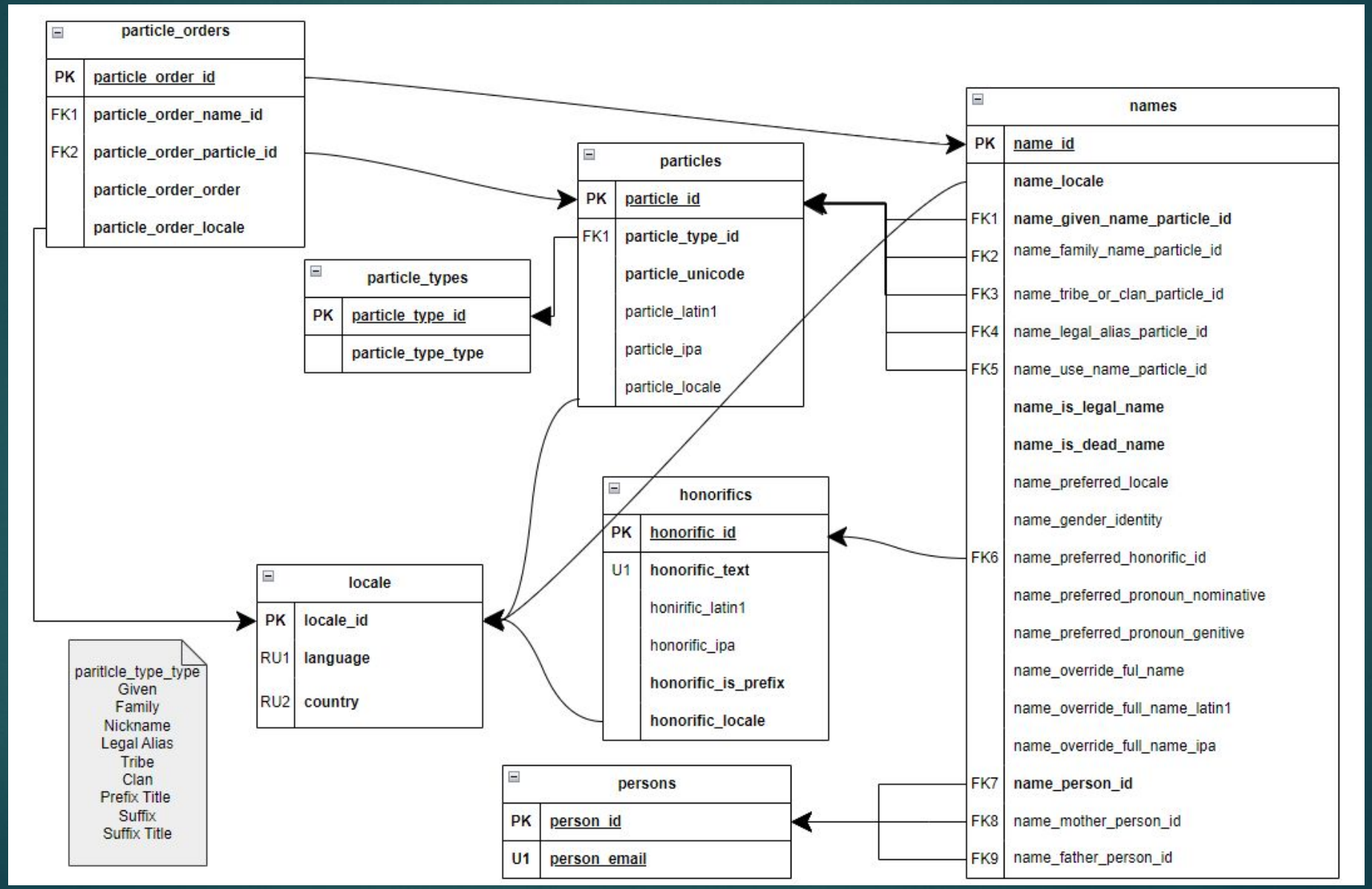
- La Monte H.P. Yarroll
- Oluwaseyi Durosinmi-Etti
- Patrick Le
- Christopher A. Murphy

- IST659, Spring 2024

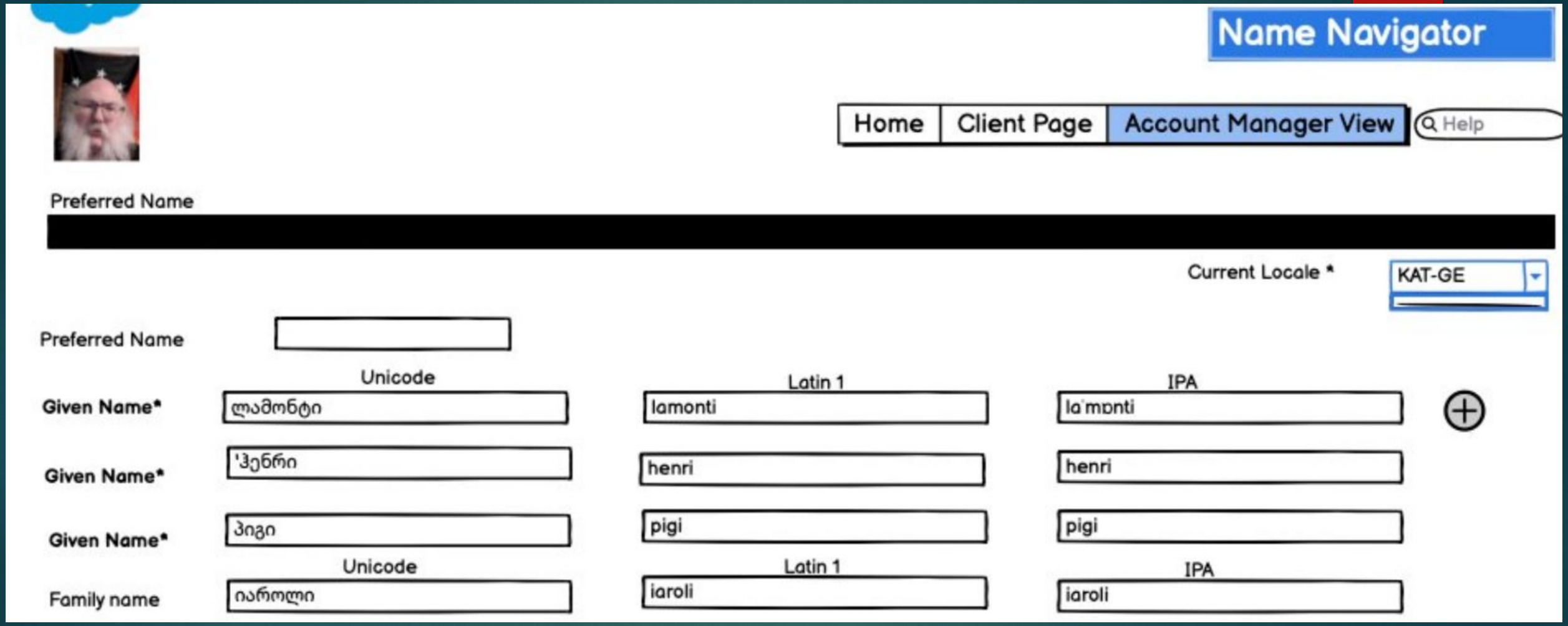
Conceptual Model



Logical Model



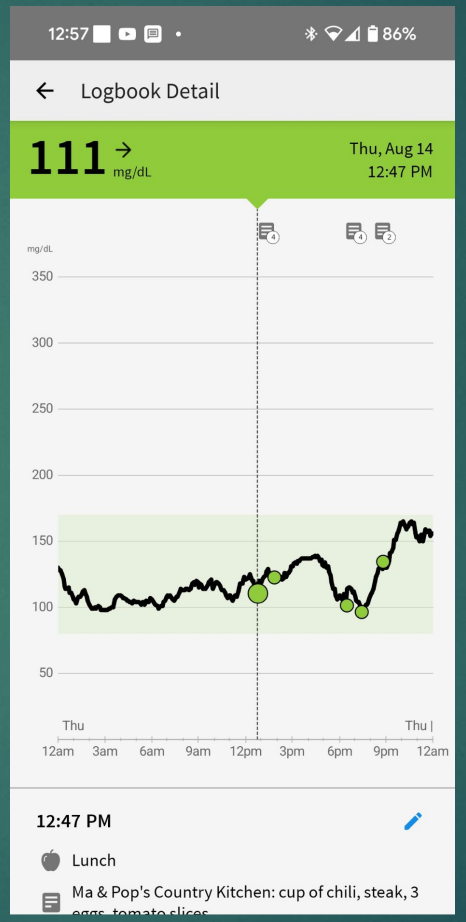
UI Wireframe



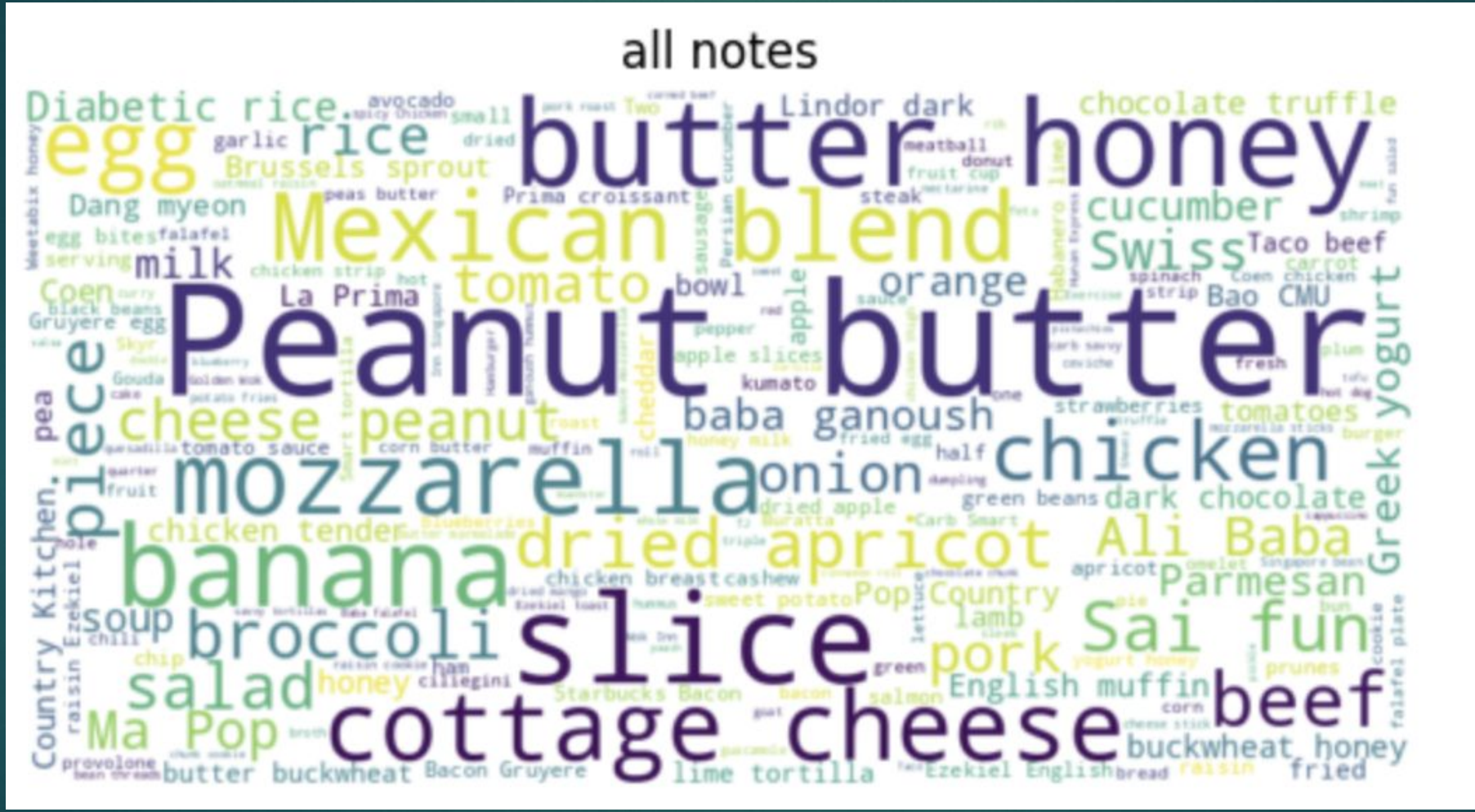
Sugarbowl CGM Analyzer

- JEN BALASI
- LA MONTE YARROLL

What is a Continuous Glucose Monitor?

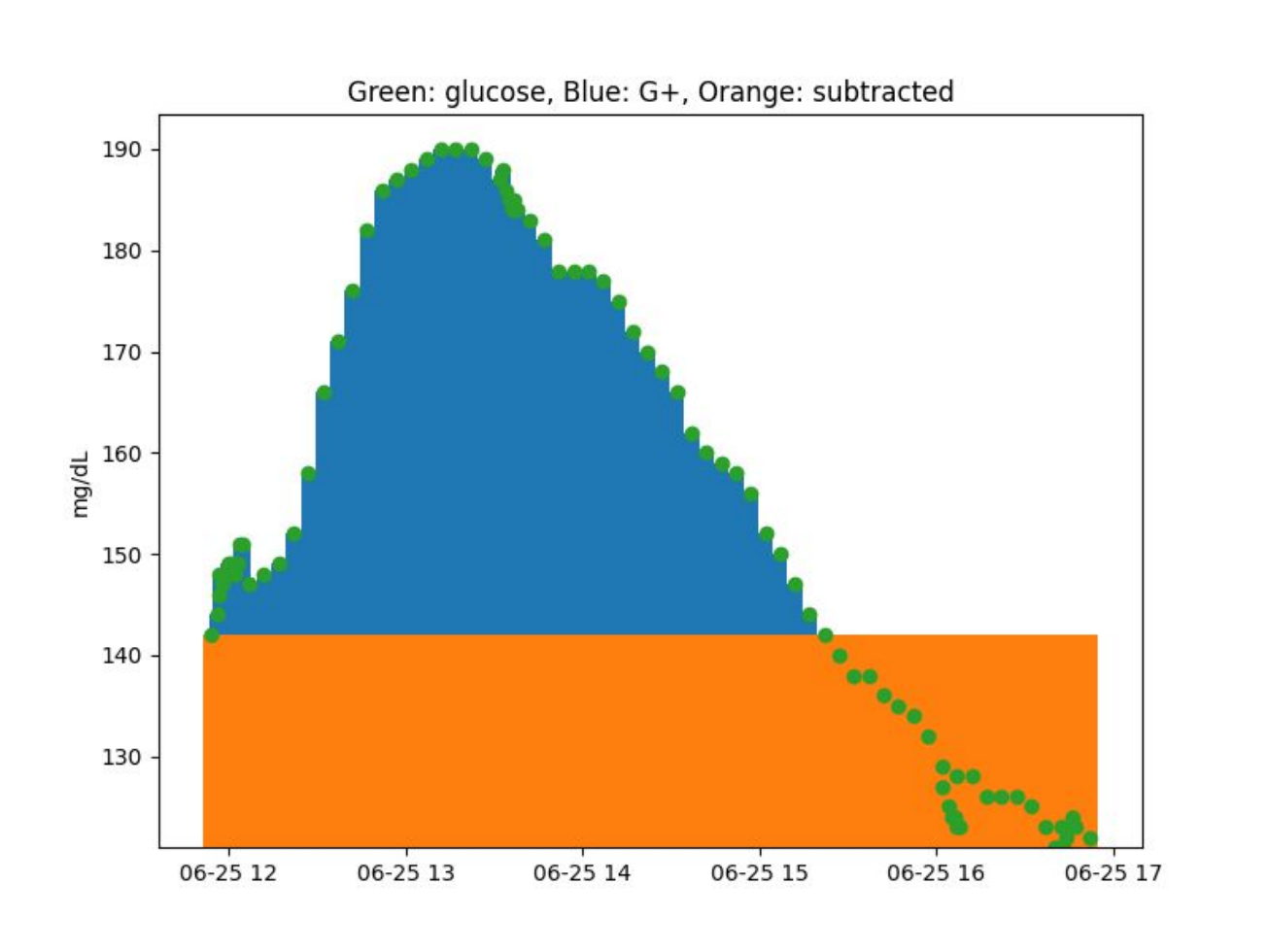


Initial Exploration EDA



Glycemic Increment

Glycemic Increment (G+) is new for our study. It's the area under the glucose curve for 2 hours from eating, minus the starting glucose. The unit is mg-hours/dL.



Glycemic Increment Results

	coef	features
0	-147.239743	apricots banana
1	-73.633590	chicken tenders donut
2	-73.633590	tenders donut
3	-52.462827	prunes dried apricots
4	-43.909830	apple banana
...	...	...
8005	56.282346	apple plum
8006	73.922804	donut
8007	92.962918	croissant
8008	117.061447	slices raisins peanut
8009	199.154914	dried apricots banana

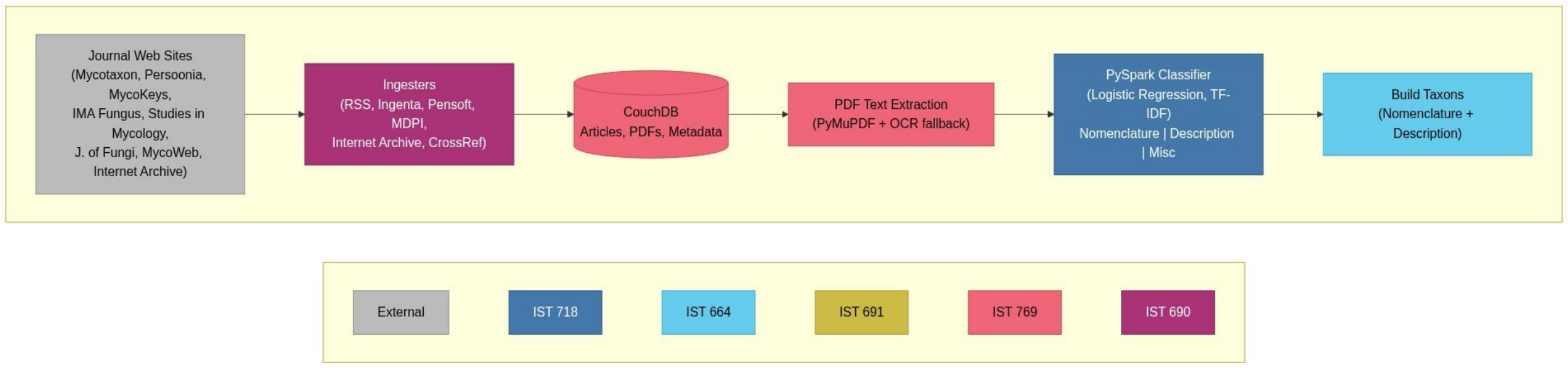
	coef	features
3767	-0.005447	peanut butter cashews
3768	-0.005447	cashews prunes
3769	-0.005447	butter cashews
3770	-0.005447	butter cashews prunes
3771	0.011467	with peach preserves
3772	0.011467	cheese with peach
3773	0.011467	cottage cheese with
3774	0.011467	with peach

Synoptic Key of Life (SKOL)

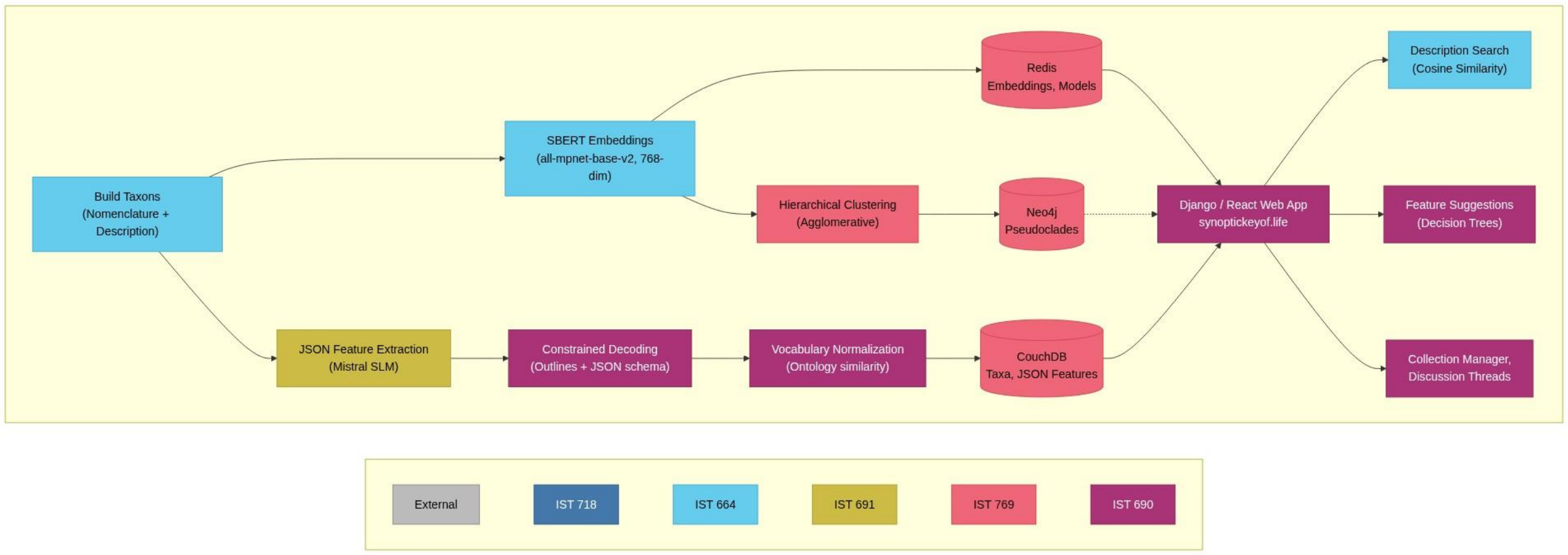
Opening the Mycological Literature to the Amateur

- La Monte H.P. Yarroll
- Christopher Murphy
- Jennifer Balasi
- Shintaro Osuga
- David Caspers
- Padmaja Kurumaddali
- Patrick Le

SKOL Architecture by Implementation Class (1 of 2)



SKOL Architecture by Implementation Class (2 of 2)



Search Page (showing JSON Feature Tool)

Welcome, **niemand** | My Collections | Help | Feedback | Contact | Account | Logout

SKOL Semantic Search α

Search taxonomic descriptions using semantic similarity

Try this:

```
campanulate pileus, stipe stains blue when cut
```

Active Collection: inat stereum × + New

ID: 601199613 View Details

Nomenclature: Unknown

Description Text (linked to collection) Flag

orange margin; hyphaesystem ketoatocalhyphae thickened walls; spores (7-) 10.5 - 11.3 (-12) × (6.5-) 9.5 - 10 (-11) µm, Q: (1-)1.1-1.1(1.1); paraphyses (100-) 116 - 137 (-147) µm;

Vocabulary | Text Features | JSON Features | Metrics

100 taxa, 739 features, depth 10

- [ ] conidia cylindrical shape 0 cylindrical
- [ ] growthform 11 description 0 epihymenium colourless, faintly yellowi
- [ ] meissner hydnoidea arrangement 0 opposite or alternate
- [ ] conidioma structure 1 1–4 mm diam.

Search Results

Search | Leucocoprinus | List Names

▸ Discussion (0)

External Identifiers

GenBank 123456 Delete

Select type... | Identifier value (e.g., 336010515) | Add

Found 22 results — From collection search: "pileus form umbo low central umbo; pileus microstr..."

Loading 22 results from collection... — From: pileus form umbo low central umbo; pileus microstr...

Leucocoprinus tenellus Pegler [nom. illegit., non (Boud.) Locq.] Context 97.3%

Label Collection

Feed: User Collection

pileus form umbo low central umbo; pileus microstructure pileipellisform cutis;asci spores per ascus 4;spores (11-) 13 - 13 (-14) × (9.5-) 10 - 10 (-11) µm, Q: (1.1-)1.3-1.3(1.4); Context

Show Full JSON

Leucocoprinus tenellus Pegler [nom. illegit., non (Boud.) Locq.] from the Context 82.6%

Label Collection

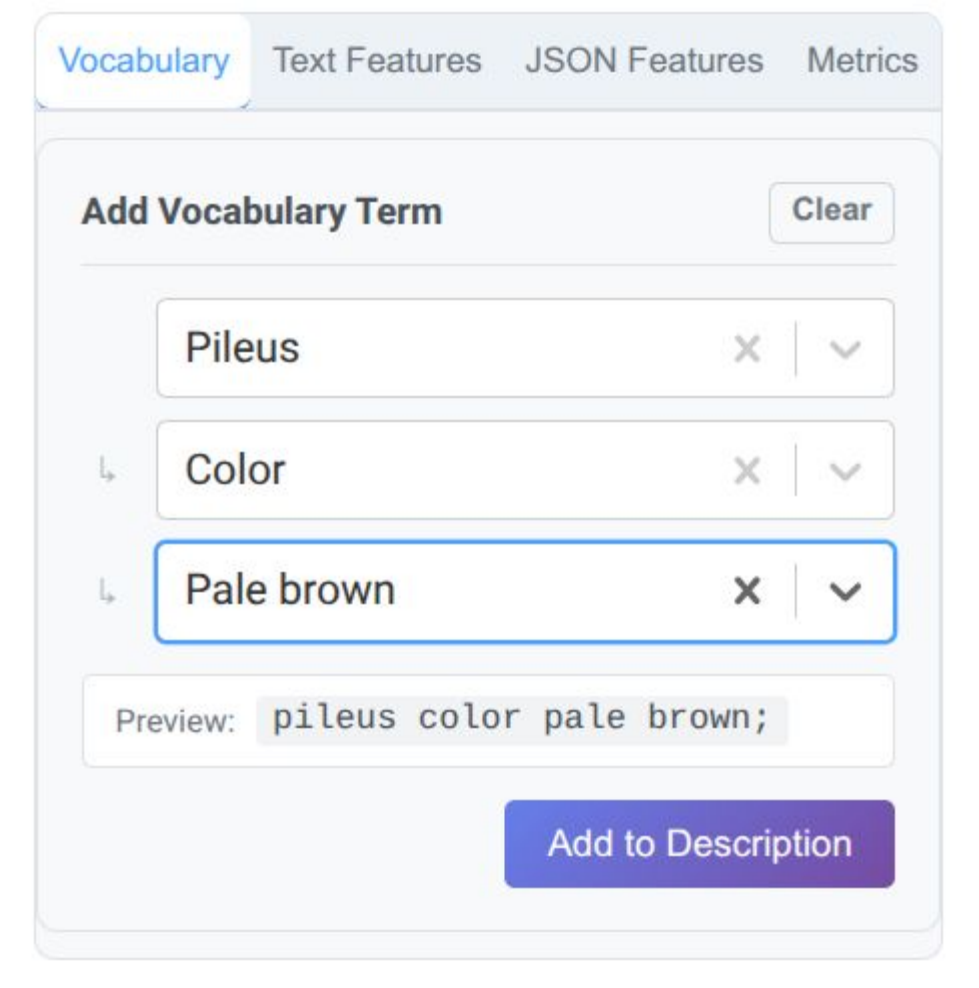
Feed: CouchDB Taxa PDF Page: 7

4b. Pileus white to yellowish-brown, pale brown or greyish; stipe whitish, Context
without reddish, greenish or bluish tinges; spores 5–8 µm in broad. 5
5a. Pileus 15–25 mm in diam., hemispheric to convex, pale brown with

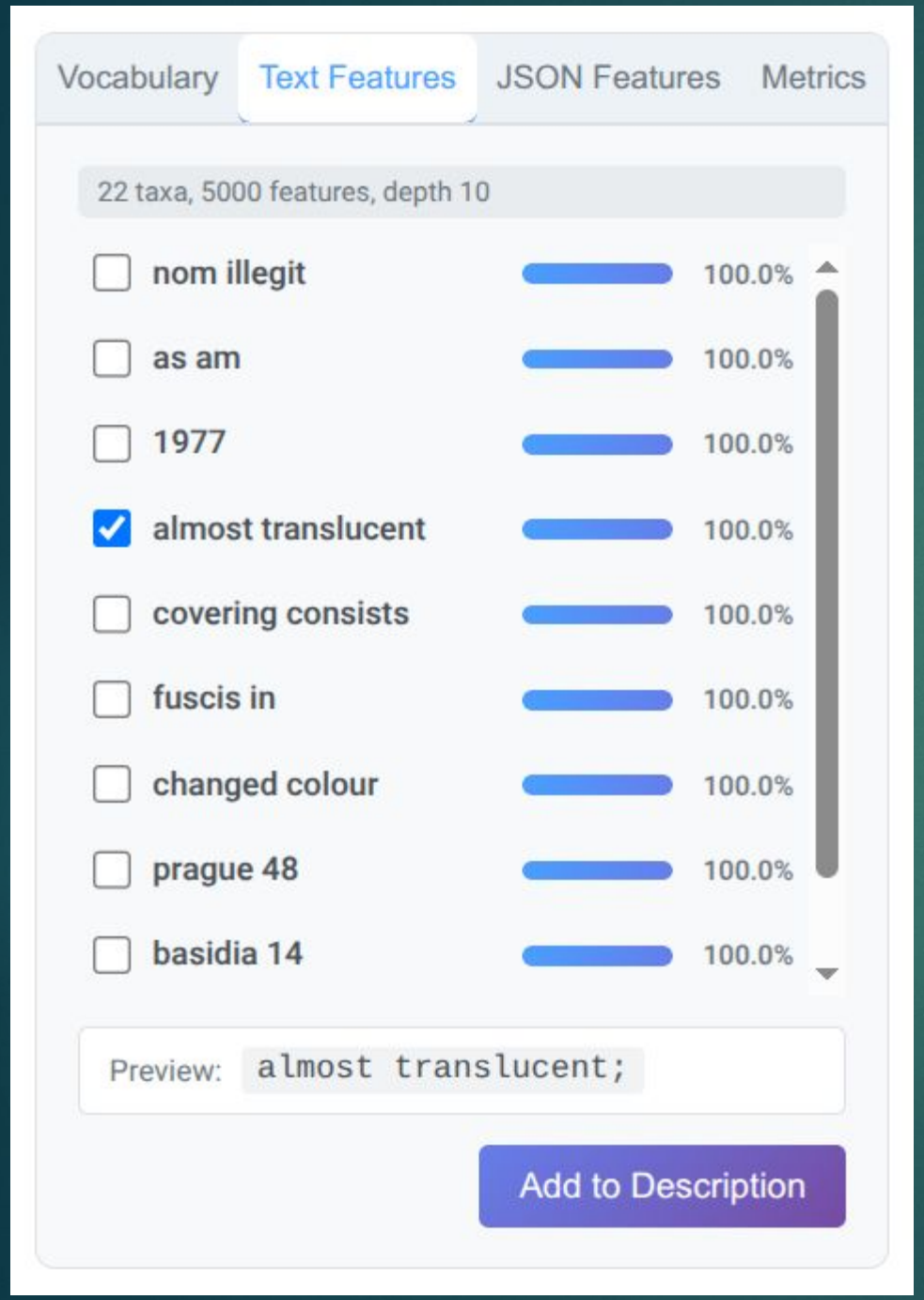
stipe greyish white to yellowish, smooth or covered by

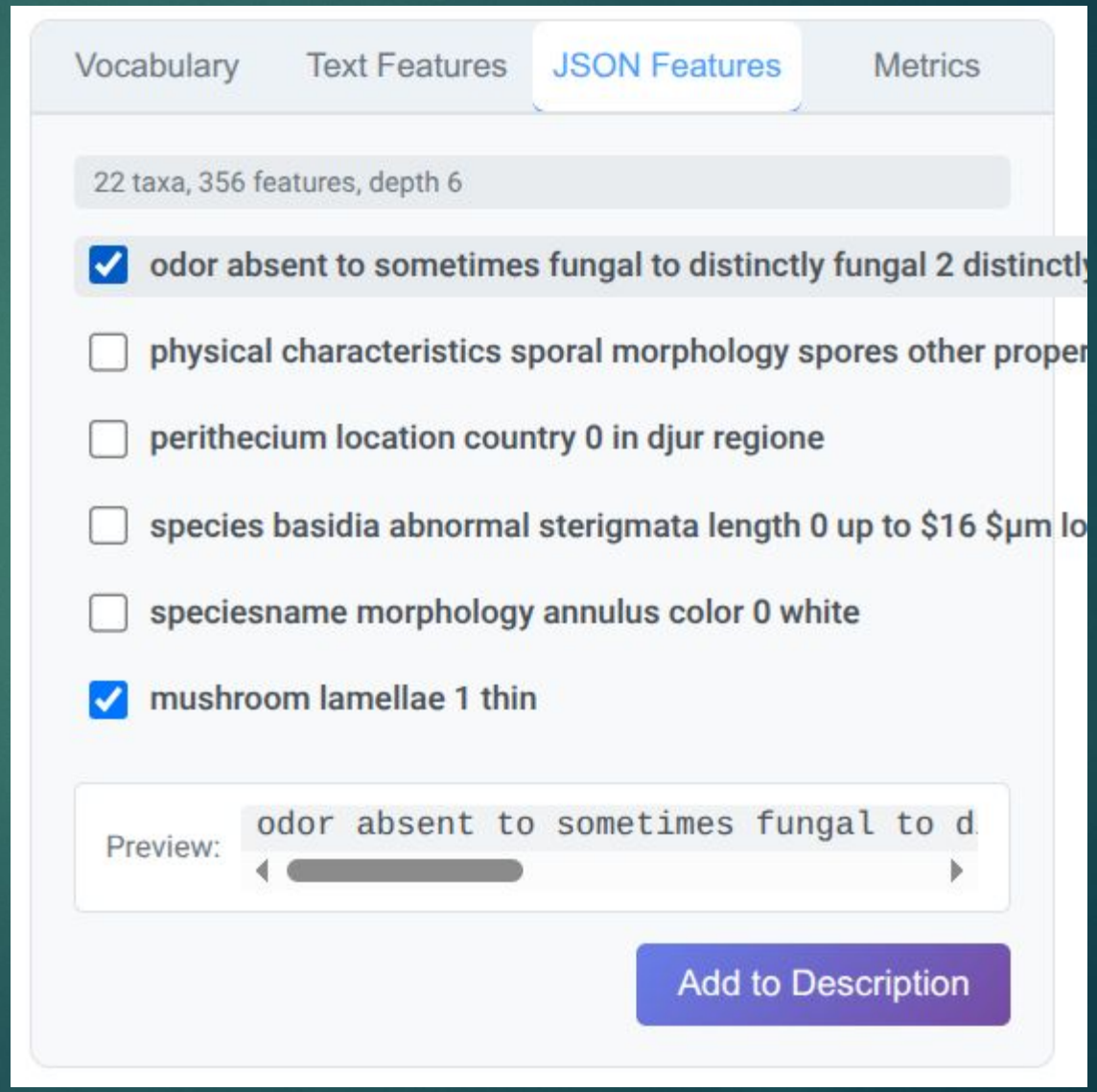
View PDF | View Source | Show Full JSON

Vocabulary Feature Tool

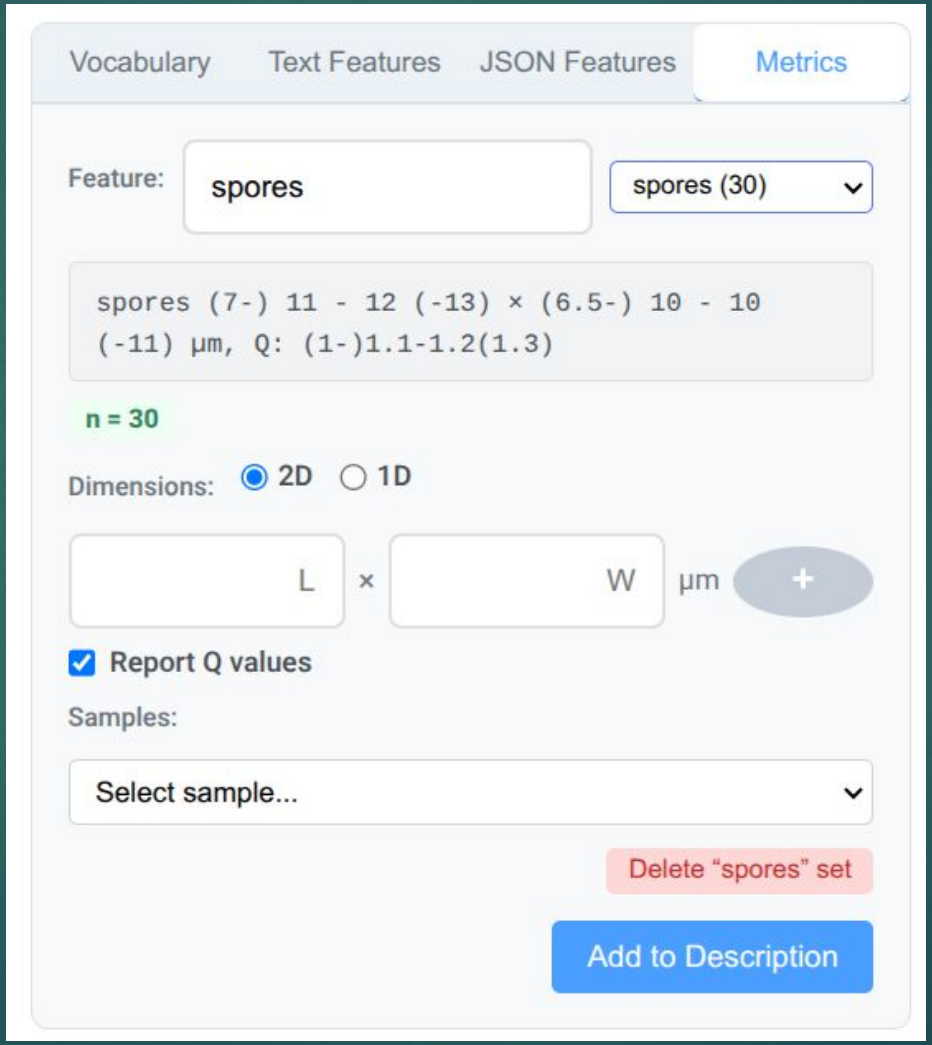


Text and JSON Feature tools

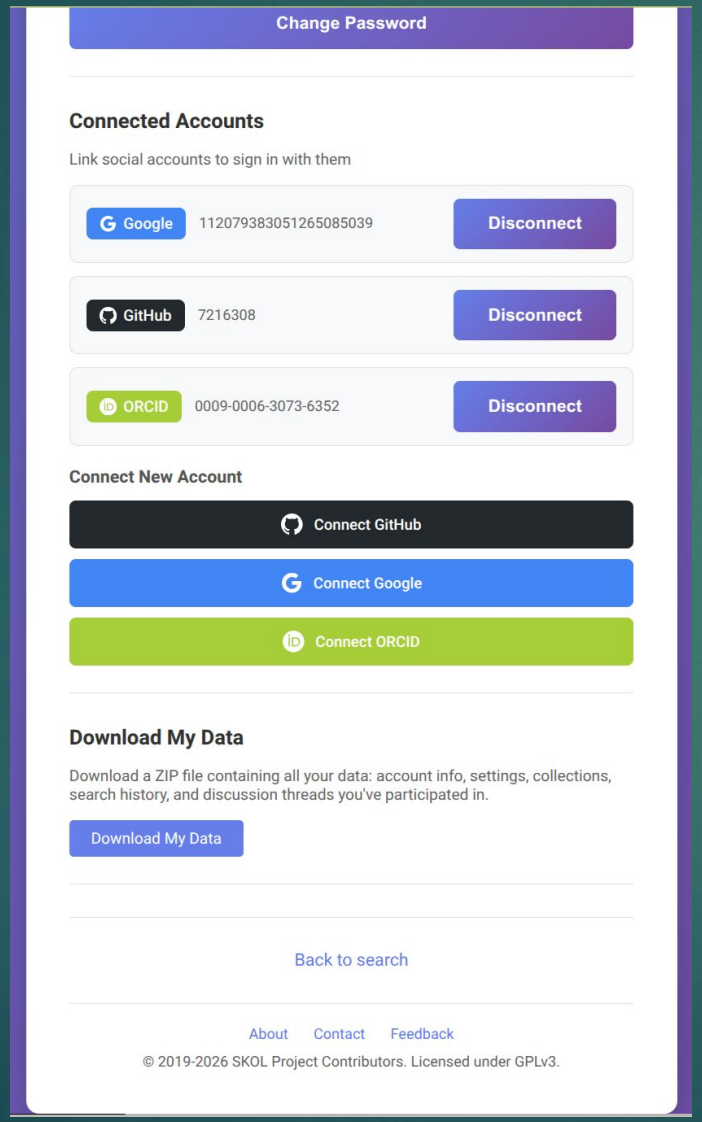


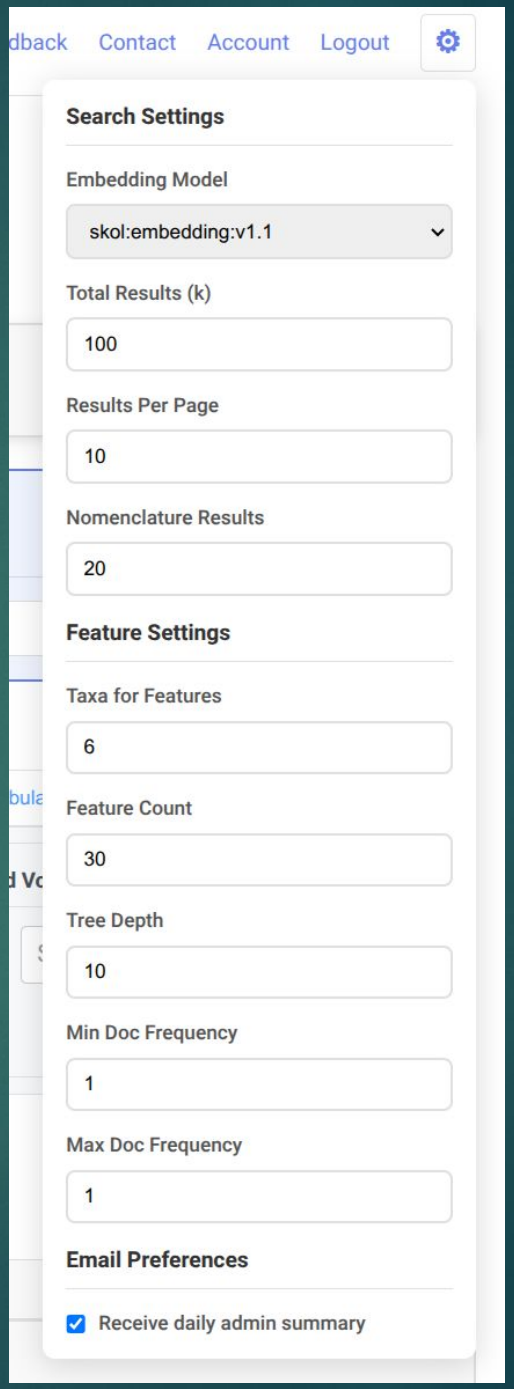


Metrics Description Tool



Account and Settings





These slides:
https://tinyurl.com/ykqwgdlk
The accompanying paper:
https://github.com/piggyatbaqaqi/skol/blob/main/IST782/portfolio2.pdf

Name Navigator module:
https://github.com/piggyatbaqaqi/names
Sugar Bowl and Glycemic Increment (G+):
https://github.com/piggyatbaqaqi/sugarbowl
The SKOL web site and source code:
https://synoptickeyof.life/skol
https://github.com/piggyatbaqaqi/skol